Vitamin A deficiency in Children

Vitamin A deficiency (VAD) is a serious health problem among children in many developing or low-income countries. In fact, 250 million children across the world suffer from VAD. It could even result in death when left untreated.

There are many animal sources (liver, whole milk, cheese, eggs, and oily fish) and plant sources (colorful fruits and vegetables as well as greens) of vitamin A. When we don't eat enough of these foods, VAD sets in, leading to various diseases. VAD can also be triggered by inefficient storage, transport, and absorption of vitamin A.

Vitamin A deficiency leads to xerophthalmia which covers a wide range of eye problems like

a) Night blindness is an early symptom of Vitamin A deficiency. When vitamin A is deficient, the formation of rhodopsin is impaired giving rise to night blindness. Children with night blindness are unable to see in poor light condition /darkness.

b) Bitot's spot- Bitot's spots are whitish deposits on the thin lining of the lower eyelid. They are usually spotted in children aged 3–6 years.



Fig 3.3 Bitot's spot

Without sufficient vitamin A, the conjunctiva becomes dull and dries (conjunctival xerosis). This leads to the buildup of keratin protein in the eye, manifesting as dry, triangular lesions with a foamy appearance called Bitot's spots.

If the symptoms of Bitot's spots are not addressed by treating vitamin A deficiency it can affect the cornea of the eye. Cornea is the clear front portion of the eye that allows light to enter and which is partially responsible for our eyes' ability to focus. Corneal xerosis is a drying up of the eyes. With corneal xerosis, the tear glands in the eye malfunction and no longer produce tears and mucus to keep the eye surface moist.



Wrinkling and dark colouring of the conjunctiva on one side, cornea dry and dull (Xerosis)

Fig 3.4 Corneal Xerosis

The cornea then becomes dry and hazy and takes on a parched appearance. It is also more prone to infections at this stage.

Prevention of Vitamin A deficiency

Eat foods which are rich in Vitamin A. Milk, egg, fish oil etc. are rich in vitamin A. Leafy vegetables like various types of greens, vegetables like carrot and fruits like papaya and mango are good sources of vitamin A. Liver, cod liver oil, butter and ghee also provide vitamin A

Based on research conducted by National Institute of Nutrition, Hyderabad, one spoon of Vitamin A syrup to children of 1-5 years age group once in 6 months also prevents vitamin A deficiency to certain extent.

When the children are given Vitamin A syrup once in six months up to five years of age, Vitamin A gets retained in the liver and is available in sufficient quantities till the next

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